

Software Requirements Specification

Ostomy Patient Management Application (v1.0)

Prepared by: Steven E. Waldren, MD, MS

1. Executive Summary & Product Vision

This application is a dual-platform (Mobile & Web) patient-facing diary designed to help ostomy patients track and manage stoma output, fluid intake, meals, and medication administration. The primary market differentiator is its ability to seamlessly integrate input, output, and medication timing to inform both the patient and their healthcare providers on the efficacy of the current care plan. While version 1.0 operates as a standalone patient diary, the architecture is designed from day one to support future electronic health record (EHR) interoperability (via FHIR) and advanced AI/predictive capabilities.

2. Platforms & Accessibility

- **Cross-Platform Availability:** The application must be fully functional and accessible via a native/hybrid mobile application (iOS and Android) as well as a standard web browser.
- **Offline-First Architecture:** Because patients will track data on-the-go (e.g., in public restrooms or while traveling), the application must utilize a local database (e.g., SQLite or Realm) to allow continuous data entry and historical review without an active internet connection. Data will automatically synchronize with the central secure cloud server once connectivity is restored.

3. Core Features & Patient Workflows

3.1. Data Entry: Intake, Output, and Meals

- **Volumetric Tracking (mL):** Patients must be able to log output and fluid intake precisely in milliliters (mL). Normal output varies by ostomy type, making exact volume tracking critical to identifying dehydration risks.
- **"Measured" vs. "Estimated" Flag:** Every volumetric entry must include a mandatory toggle or flag indicating whether the volume was strictly measured (e.g., using a graduated pitcher) or estimated by the patient.
- **Chart by Exception & Dynamic Widgets:** To minimize data entry friction, the application will analyze the user's latest entries to determine common patterns. It will generate dynamic "Quick-Add" widgets on the dashboard. The user can tap a widget to log it instantly or open it as a pre-filled template to edit specific fields.

3.2. Medication Management

- **Targeted Medication Focus:** The medication logger will focus primarily on treatments that manage ostomy symptoms (e.g., anti-diarrheals for high-output, hydration supplements, or barrier creams).
- **RxNorm Integration & Patient-Friendly Terminology:** The backend will utilize the RxNorm database to ensure standardized nomenclature for clinical drugs. The UI will leverage a patient-friendly interface terminology mapping so patients can easily search for the correct medication without complex clinical jargon.
- **Timestamping:** Precise recording of medication administration times is required to correlate drug efficacy with stoma output spikes.

3.3. The Physician-Focused View (Care Plan Efficacy)

- **Daily Net Fluid Balance:** A core metric prominently displayed, calculating total fluid intake minus total stoma output (and other recorded losses) to instantly highlight dehydration trends.
- **Chronological Correlation Visuals:** Data will be visualized across time periods (Morning, Afternoon, Evening, Night). Line or bar graphs will overlay medication administration times and fluid intake directly against stoma output volumes.
- **Outlier & Anomaly Highlighting:** The report will automatically flag anomalies (e.g., output exceeding 1,000 mL/day while intake remains low) to focus the physician's attention on actionable care plan changes.

4. Data Model & Interoperability Strategy

- **FHIR Mapping Readiness:** Output and intake logs will be structured to map cleanly to the FHIR Observation resource.
- **Method Attributes:** The "Estimated vs. Measured" distinction will be stored as an explicit Observation.method attribute to ensure clinical validity when data is eventually transmitted to a provider.
- **Medication Data:** Medication records will be stored using their unique RxNorm Concept Unique Identifiers (RXCUIs) to guarantee seamless semantic interoperability with provider EHRs in the future.

5. Security & Compliance

- **HIPAA Compliance:** The application will adhere to full HIPAA standards from day one, including encryption of patient health information (PHI) at rest and in transit (TLS 1.3+).
- Secure authentication protocols (OAuth 2.0 / OpenID Connect) with biometric login support for mobile.
- Automated session timeouts and execution of Business Associate Agreements (BAAs) with cloud providers.

6. User Stories

Epic 1: Authentication, Security & HIPAA Compliance

- **As a user**, I want to securely log in using biometric authentication (FaceID/TouchID) on my mobile device so that I can access my diary quickly without compromising my health data.
- **As a user**, I want my active session to automatically time out after a period of inactivity so that my health information remains secure if I leave my device unattended.
- **As a system administrator**, I need all Protected Health Information (PHI) to be encrypted at rest and in transit so that the application fully complies with HIPAA standards.
- **As a user**, I want to be able to reset my password securely using a multi-factor authentication (MFA) process so that I can regain access to my account safely.

Epic 2: Data Entry & Logging (Intake, Output, & Meals)

- **As a patient**, I want to log my stoma output volume in exact milliliters (mL) so that I can accurately track my fluid loss.
- **As a patient**, I want to toggle a flag indicating whether my logged output was "Measured" or "Estimated" so that my physician understands the precision of the data point.
- **As a patient**, I want to log my daily fluid intake in milliliters (mL) so that I can track my hydration against my output.
- **As a patient**, I want to log the meals I eat, including a timestamp, so that I can see how specific foods affect my stoma output later in the day.
- **As a data architect**, I need the database to store output and intake logs in a structure mapped to the HL7 FHIR Observation resource so that the data is ready for future EHR integration.

Epic 3: Smart Dashboard & Quick-Add Widgets

- **As a patient**, I want the dashboard to display dynamic "Quick-Add" widgets based on my most frequent recent entries so that I can log routine events with a single tap.
- **As a patient**, I want the ability to tap a Quick-Add widget and open it as a draft so that I can make minor adjustments to the pre-filled data before saving.
- **As a patient**, I want to set custom default values for my standard cup sizes or usual fluid intakes so that I can chart by exception.

Epic 4: Medication Management & Terminology

- **As a patient**, I want to search for medications using common, patient-friendly terms so that I can easily find and log the correct symptom-management medication.
- **As a system architect**, I want the medication search function to map patient selections to backend RxNorm Concept Unique Identifiers (RXCUIs).
- **As a patient**, I want to log the precise time I administered a medication so that its effect on my subsequent stoma output can be tracked.

Epic 5: The Physician-Focused View & Reporting

- **As a physician**, I want to view a calculated "Daily Net Fluid Balance" (total intake minus total output) so that I can instantly assess a patient's risk of dehydration.
- **As a physician**, I want to view a chronological visual overlay showing stoma output spikes plotted against fluid intake and medication administration times.
- **As a physician**, I want the report to automatically highlight data anomalies (e.g., output exceeding 1,000 mL/day) so that I can focus immediately on actionable clinical risks.
- **As a patient**, I want to generate a secure, read-only link or PDF export of the physician-focused view so that I can easily share my latest data during a clinic visit.

Epic 6: Offline-First Architecture & Synchronization

- **As a patient**, I want to log my output and meals even when my device has no internet connection so that my tracking remains uninterrupted.
- **As a patient**, I want to view my recent historical data and dashboard widgets while offline so that I can reference my daily progress anywhere.
- **As a system**, I need to automatically detect when network connectivity is restored and sync locally cached data to the secure cloud server so that the central database is always up to date.

7. Detailed Acceptance Criteria

The following details the Acceptance Criteria (AC) for Epic 2: Data Entry workflows, mapped for test-driven development.

User Story 2.1: Log Stoma Output Volume

AC 1: Numeric Validation

- **Given** the user is on the "Add Output" screen,
- **When** they enter a value into the Volume field,
- **Then** the field must only accept positive integers (no decimals or negative numbers).

AC 2: Out-of-Bounds Error Handling

- **Given** the user is entering an output volume,
- **When** the user inputs a value greater than 2,000 mL,
- **Then** the system will display a soft warning prompt: *"This is a high volume for a single entry. Please confirm this amount is correct,"* before allowing the user to save.

AC 3: Timestamp Editing

- **Given** the user is logging an output event,
- **When** the screen loads,
- **Then** the Date and Time fields auto-populate with the current system time, but remain fully editable so the user can backdate an entry.

User Story 2.2: The "Measured vs. Estimated" Flag

AC 1: Mandatory Selection

- **Given** the user has entered a volume on the "Add Output" screen,
- **When** they attempt to tap "Save" without selecting an entry method,
- **Then** the system prevents saving and highlights the "Measured / Estimated" toggle group with a required field error.

AC 2: Visual Differentiation in History

- **Given** the user is viewing their daily log history,
- **When** an output entry is displayed,
- **Then** it must contain a clear visual indicator (e.g., an icon or badge) denoting whether that specific volume was Estimated or Measured.

User Story 2.3: Log Fluid Intake

AC 1: Fluid Type Categorization

- **Given** the user is logging fluid intake,
- **When** they enter the volume,
- **Then** they must be presented with an optional categorized dropdown/list for the type of fluid (e.g., Water, ORS, Coffee/Tea, Juice) to aid in dietary trigger analysis.

AC 2: Quick-Add Standard Sizes

- **Given** the user is on the "Add Intake" screen,
- **When** they view the volume input section,
- **Then** they see at least three configurable quick-select buttons for standard container sizes (e.g., 250 mL glass, 500 mL bottle) that auto-fill the volume field when tapped.

User Story 2.4: Log Meals

AC 1: Text Entry & Tagging

- **Given** the user is on the "Add Meal" screen,
- **When** logging food,
- **Then** they can use a free-text area to describe the meal AND select from optional quick-tags (e.g., High Fiber, Dairy, High Sugar).

AC 2: Meal Size Estimation

- **Given** the user is logging a meal,
- **When** saving the entry,
- **Then** they must select a relative size modifier (Small/Snack, Medium/Standard, Large/Heavy).

User Story 2.5: FHIR Mapping Readiness

AC 1: JSON Payload Structure

- **Given** an output or intake record is saved,
- **When** the API receives the payload,
- **Then** the data must be stored with keys mapping directly to FHIR R4 standard fields (e.g., `resourceType: "Observation", valueQuantity.value`).

AC 2: Method Attribute Mapping

- **Given** a user saves an output entry marked as "Estimated",
- **When** the database writes the record,
- **Then** it must correctly populate the FHIR `Observation.method` attribute using the appropriate SNOMED CT code for "Estimation technique".